

Autodesk® Scaleform®

Scaleform



Scaleform 4.0



: Matthew Doyle
: 3.0 
: 2011  25

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Autodesk® Scaleform® 4.2

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1		1
1.1		Error! Bookmark not defined.
1.2		Error! Bookmark not defined.
2		Error! Bookmark not defined.
2.1	Draw Primitives	Error! Bookmark not defined.
2.2		Error! Bookmark not defined.
2.3	Artwork	Error! Bookmark not defined.
2.3.1	vs	Error! Bookmark not defined.
2.3.2		Error! Bookmark not defined.
2.3.3		Error! Bookmark not defined.
2.3.4		Error! Bookmark not defined.
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3.1		Error! Bookmark not defined.
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3.3	Advance	Error! Bookmark not defined.

3.4 onEnterFrame and Event.ENTER_FRAME..... **Error! Bookmark not defined.**

3.4.1 onEnterFrame **Error! Bookmark not defined.**

3.5 ActionScript 2 **Error! Bookmark not defined.**

3.5.1 onClipEvent **Error! Bookmark not defined.**

3.5.2 Var **Error! Bookmark not defined.**

3.5.3 (Precaching)..... **Error! Bookmark not defined.**



3.5.4 **Error! Bookmark not defined.**

3.5.5 **Error! Bookmark not defined.**

4 HUD **Error! Bookmark not defined.**

4.1 SWF **Error! Bookmark not defined.**



4.2 SWF **Error! Bookmark not defined.**



4.3 **Error! Bookmark not defined.**

4.4 (..... **Error! Bookmark not defined.**



4.5 HUD **Error! Bookmark not defined.**

5 **Error! Bookmark not defined.**

5.1 **Error! Bookmark not defined.**

5.2 **Error! Bookmark not defined.**

6 **Error! Bookmark not defined.**

1

Autodesk® Scaleform® 4.0

Adobe® Flash®

Flash

ActionScript™

(AS)

(HUD)

HUD

HUD

Flash

Scaleform 3.1

Flash

AS

Scaleform 3.x

PC

“one size fits all” (

Best practices

(U)

. Flash Studio CS5

.FLA

ML

, AS



• Flash

0



• FLA

0



•

0



•

0

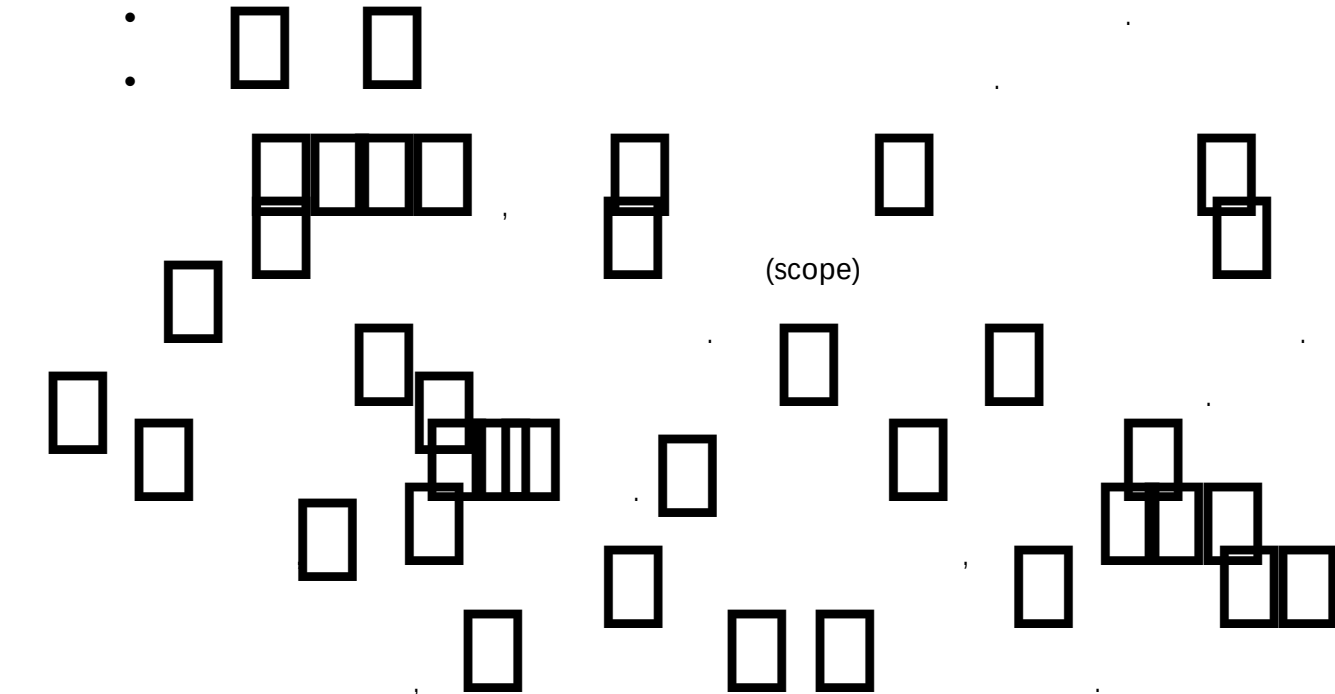


best practice



Flash





1.1

.

(

)

UI



Scaleform 3.3

Scaleform Common Lightweight Interface Kit (CLIK™)

1.2

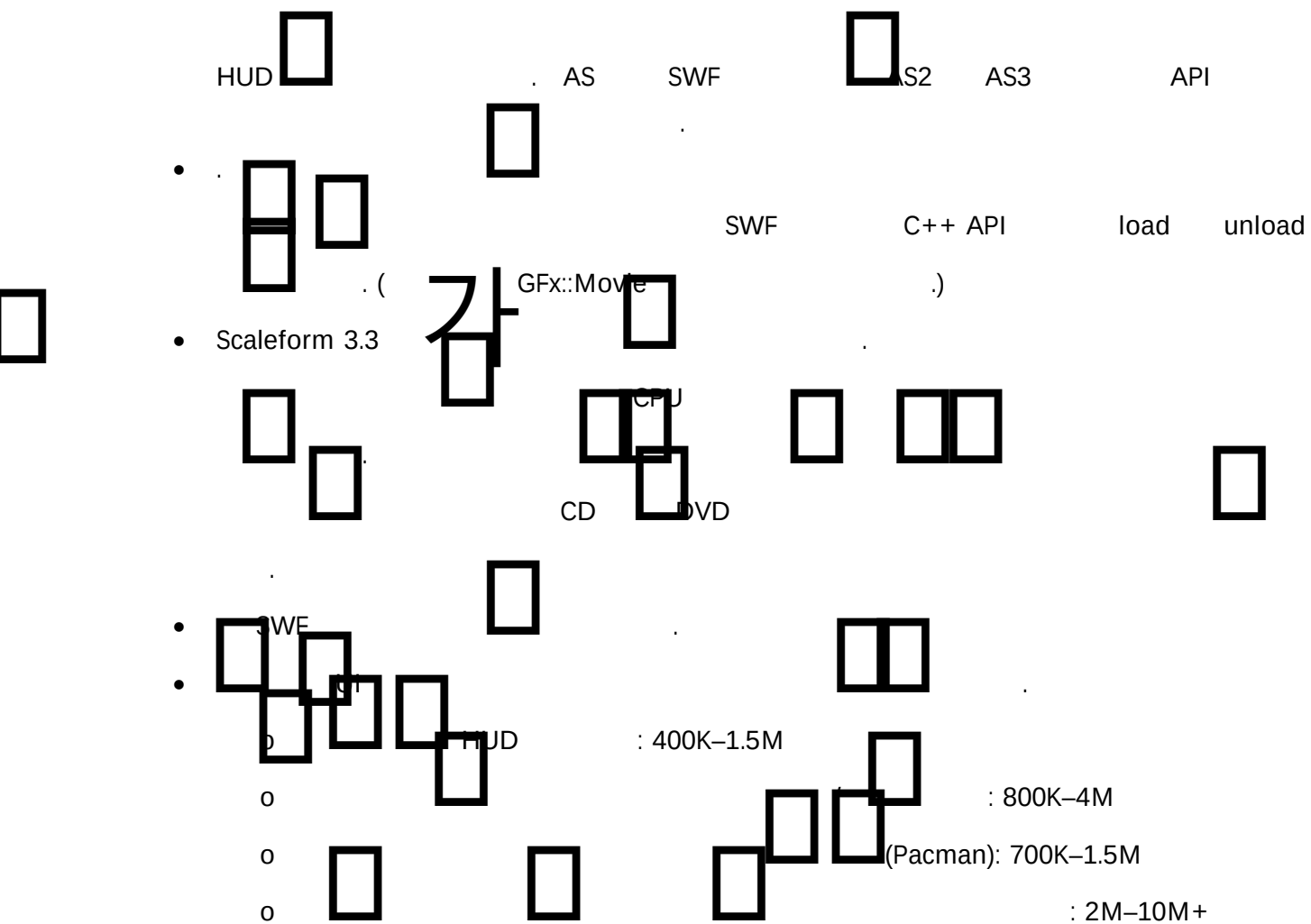
.

.

Flash

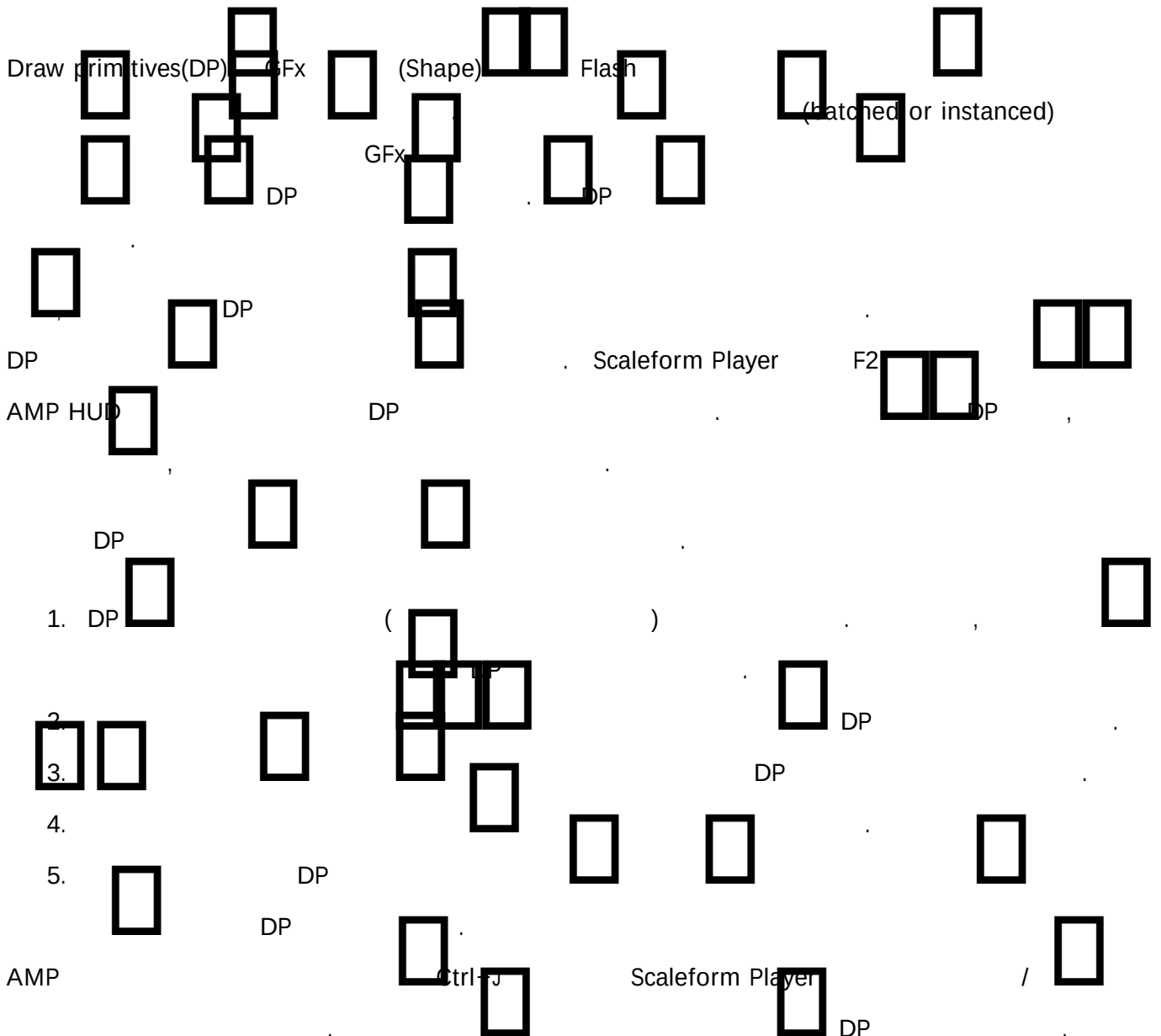
(transform)

, MMORPG



2

2.1 Draw Primitives



2.2

1.

2.

3.

variance
(testing)
`alpha=0`
`visible=false` rather than `alpha=0` (in AS3).
`stop()`

`alpha=0`

`visible=false`

(in AS2) and

2.3 Artwork

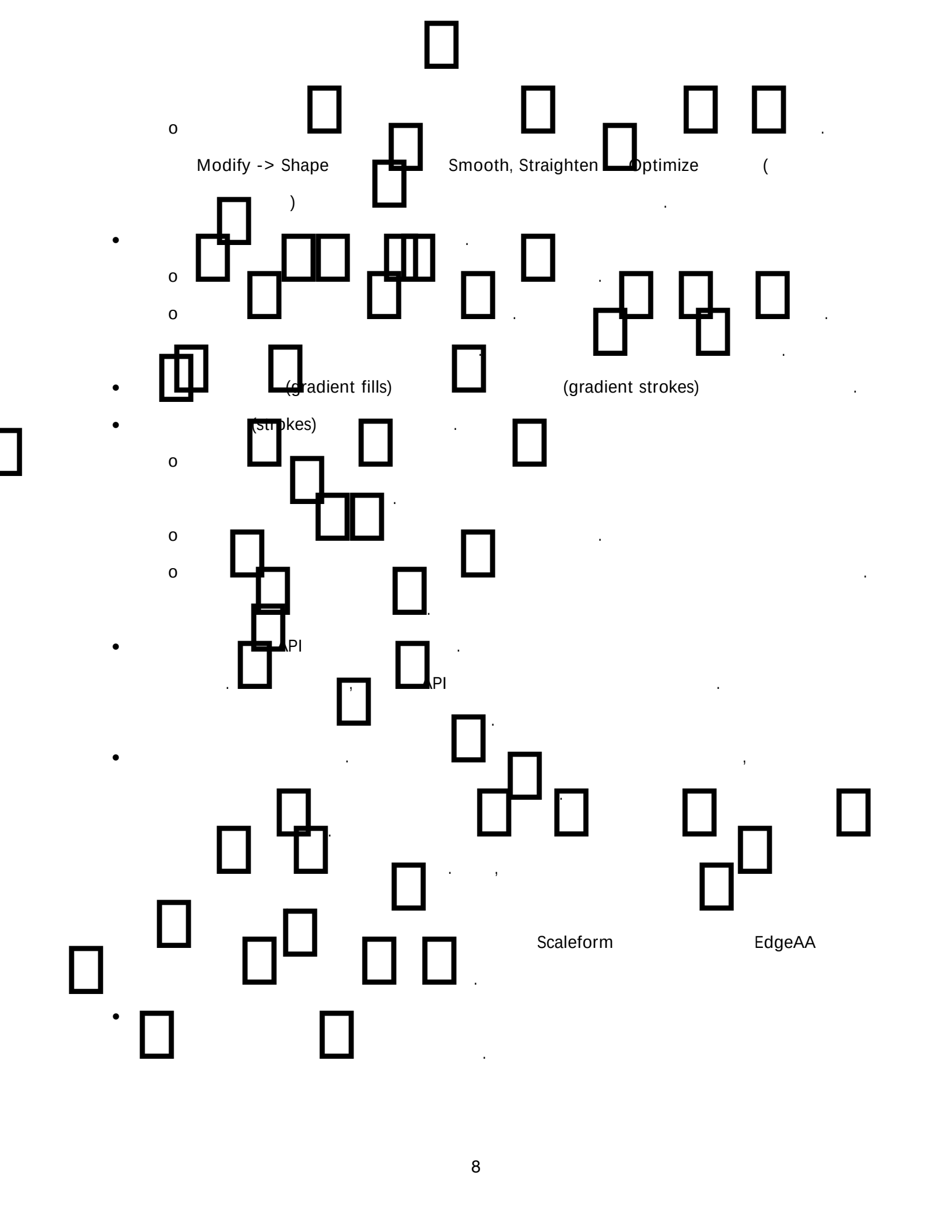
2.3.1

vs

Scaleform

2.3.2





Modify -> Shape

Smooth, Straighten

Optimize

(

)

(gradient fills)

(gradient strokes)

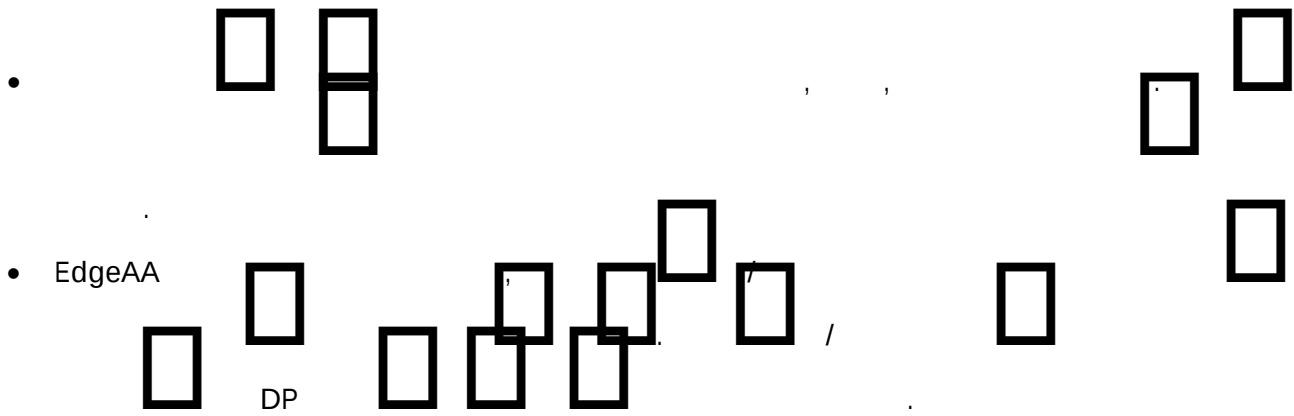
(strokes)

API

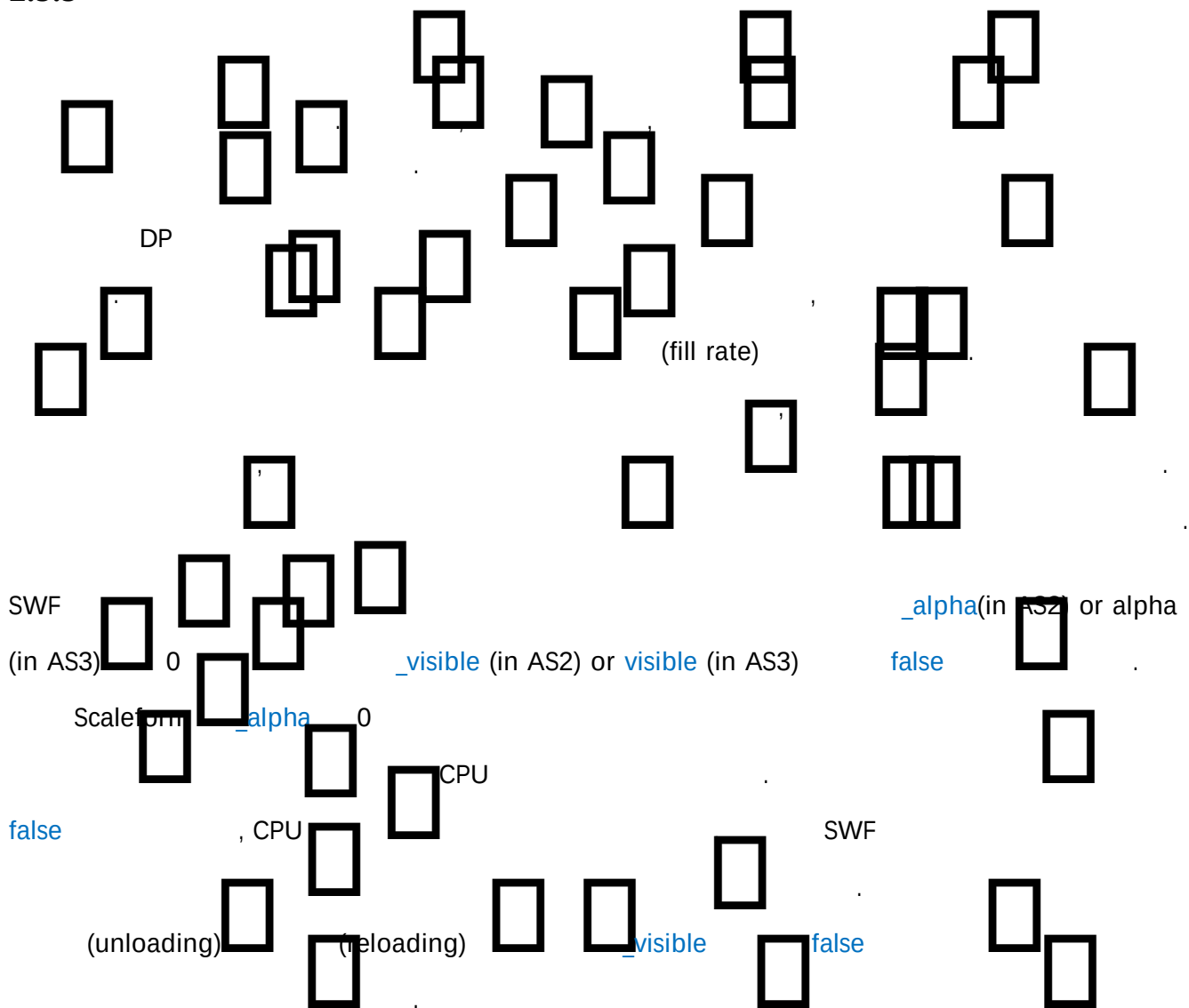
API

Scaleform

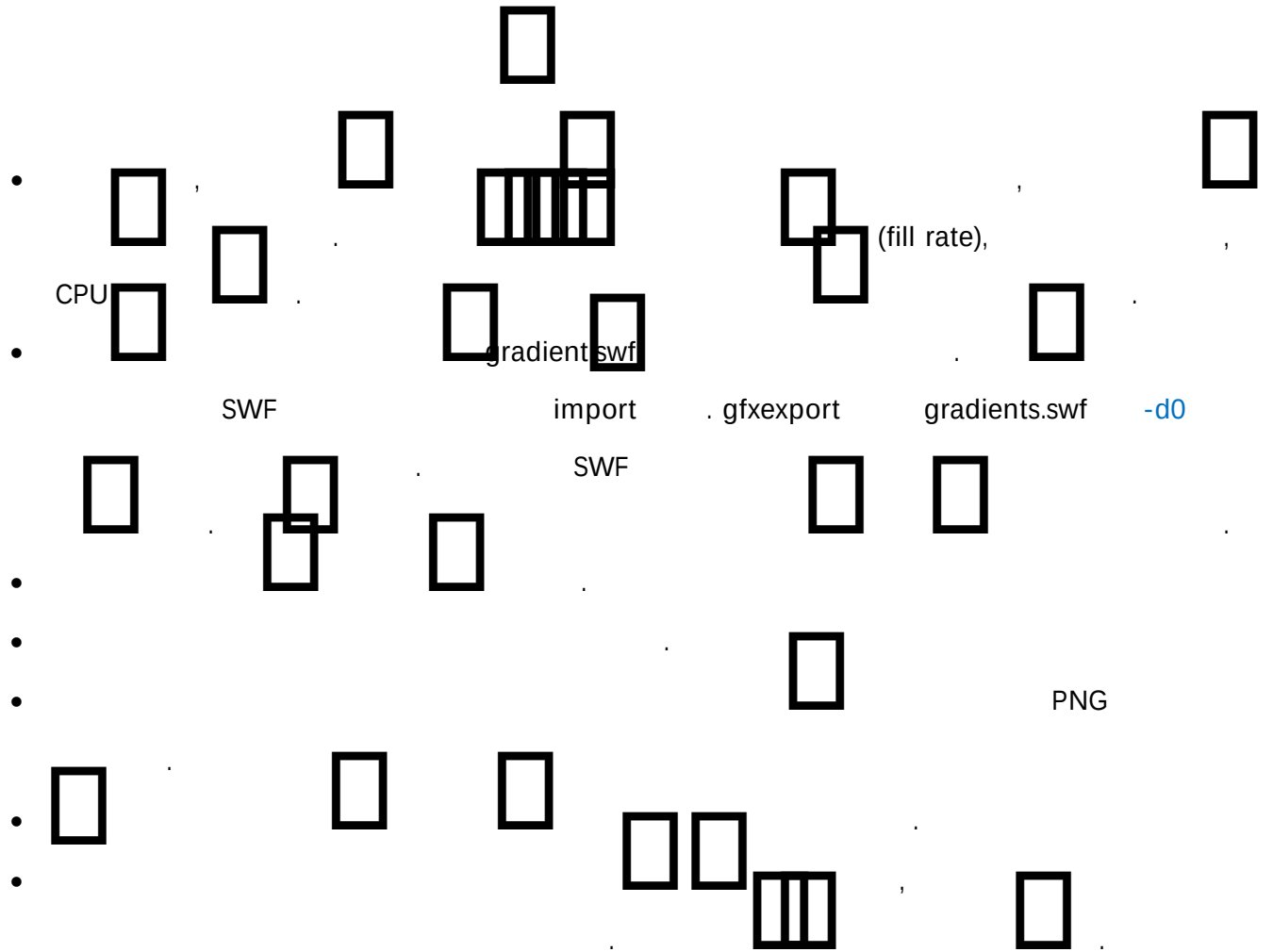
EdgeAA



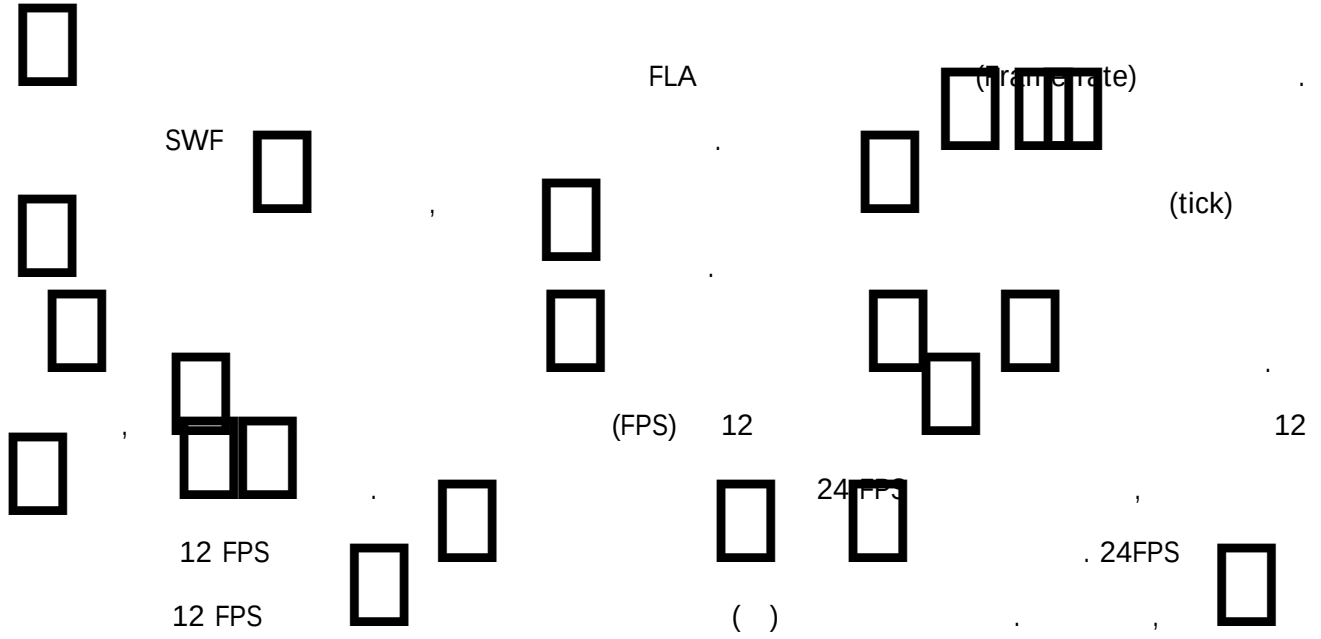
2.3.3



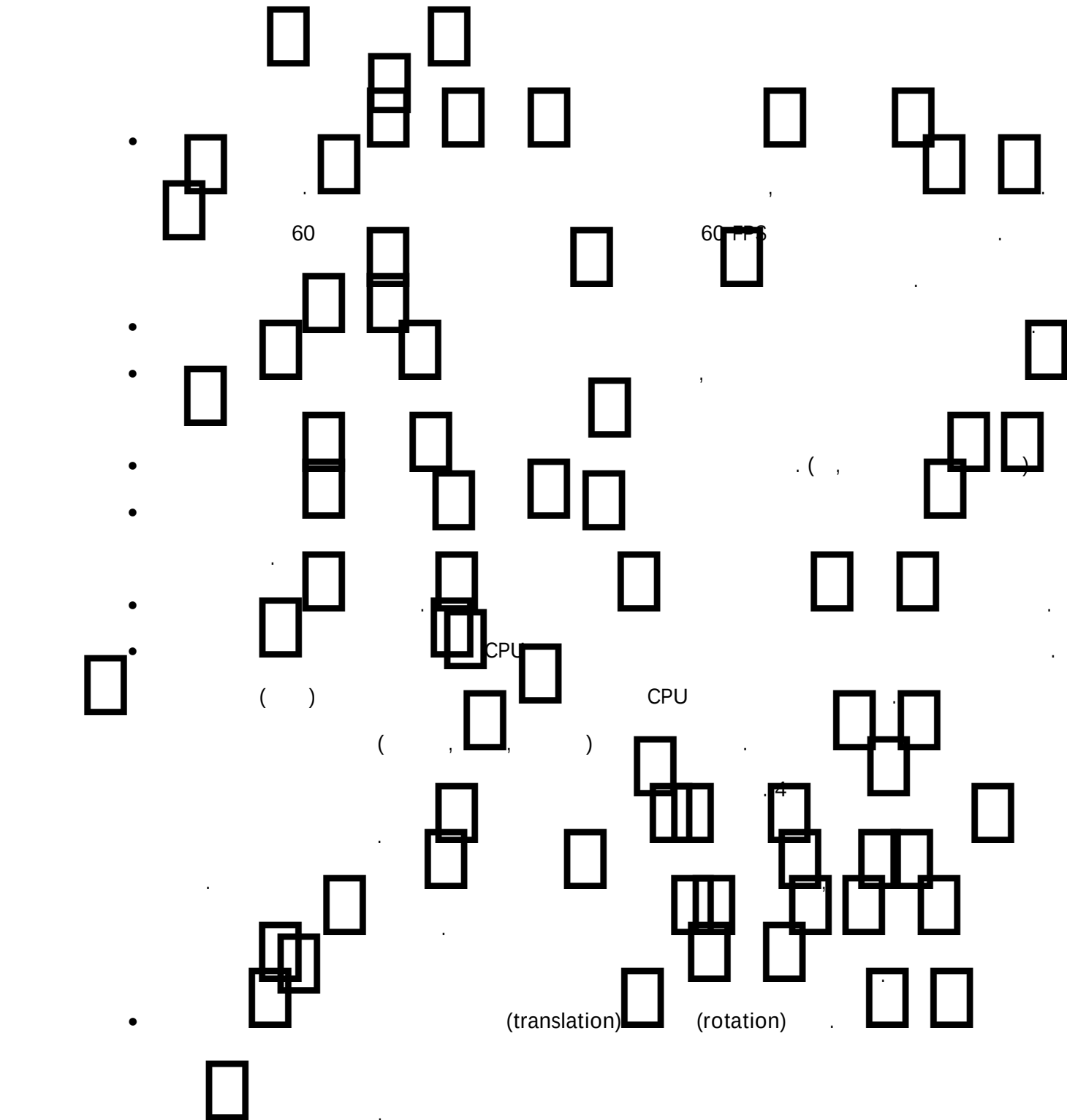
10



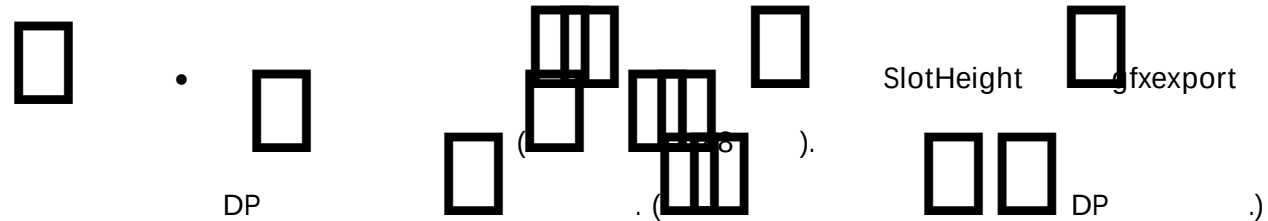
2.3.4

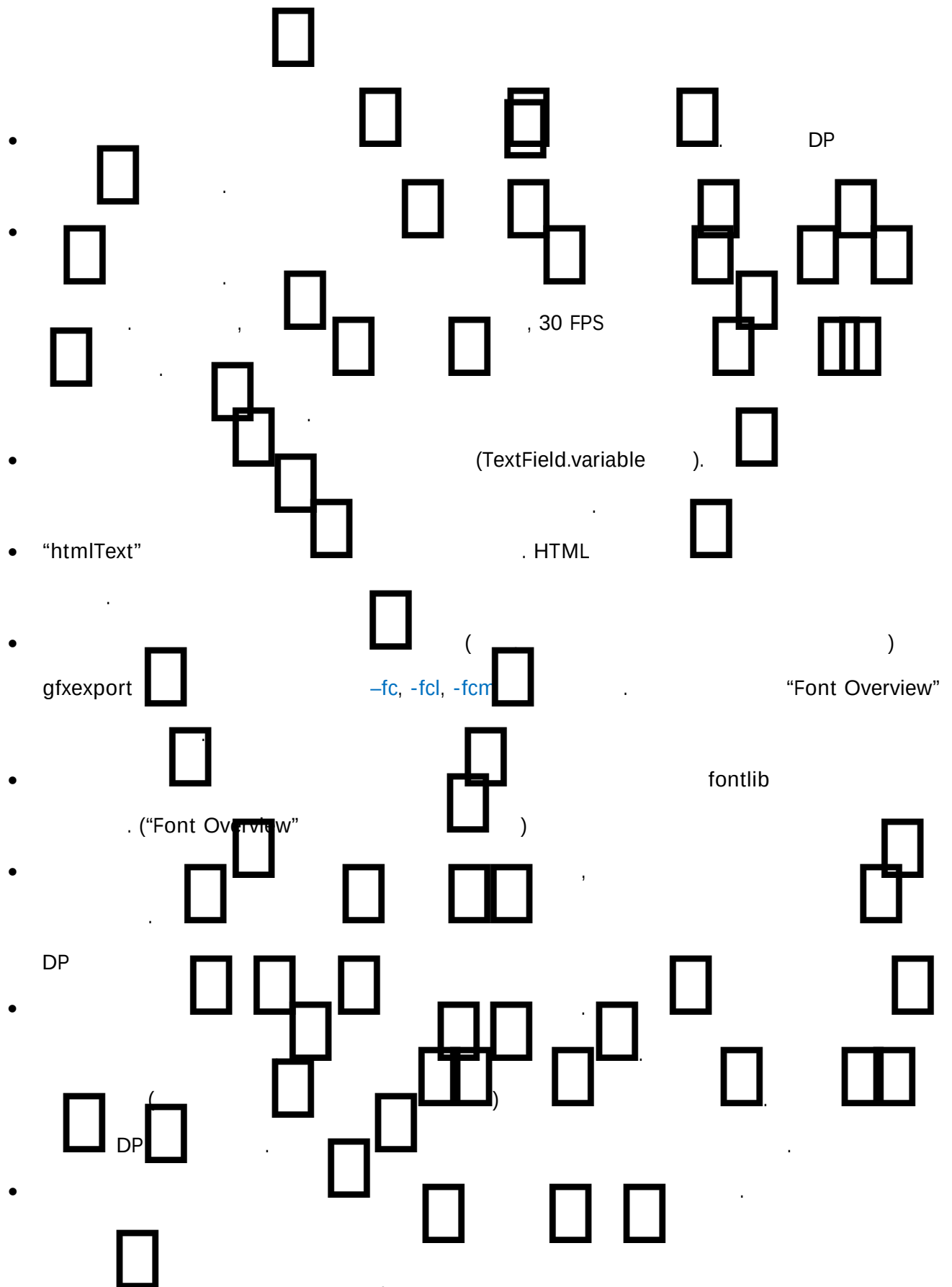


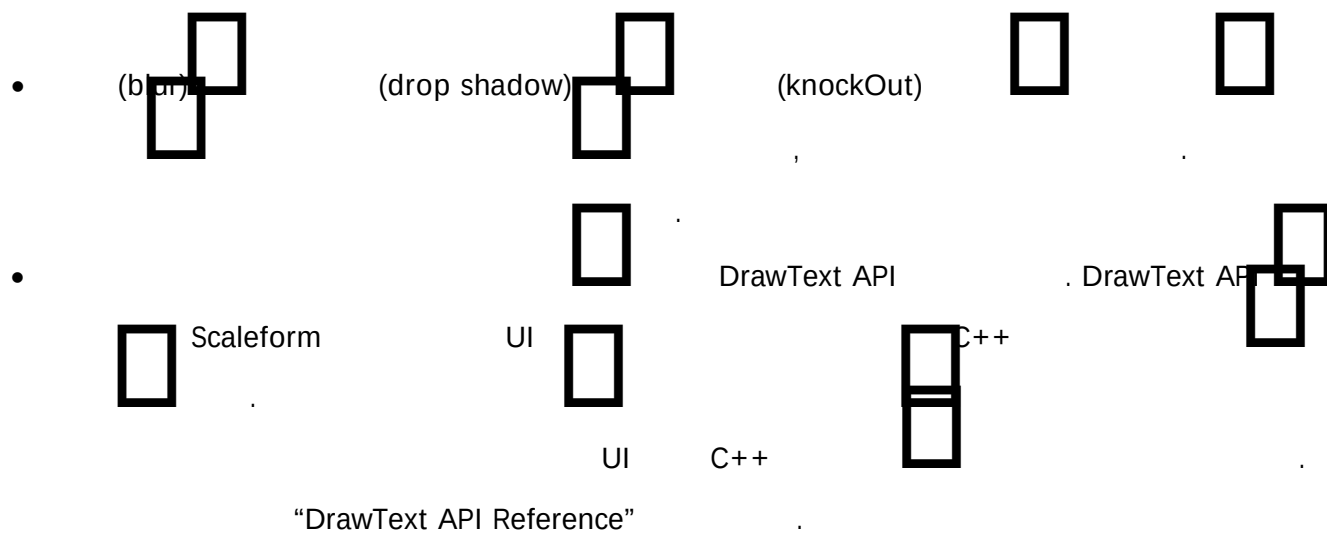




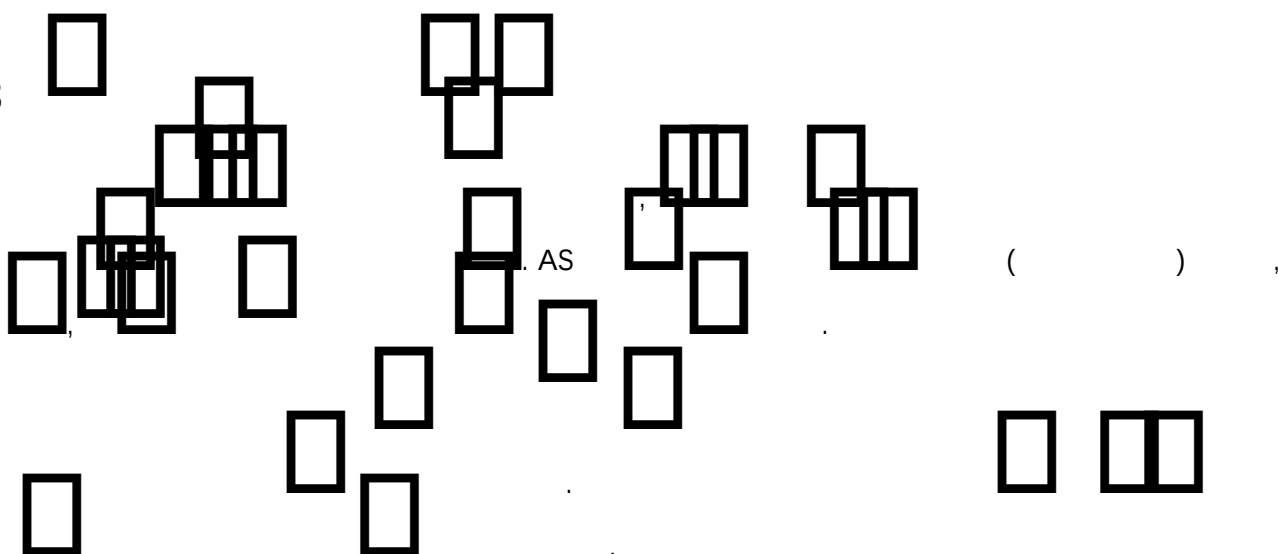
2.3.5



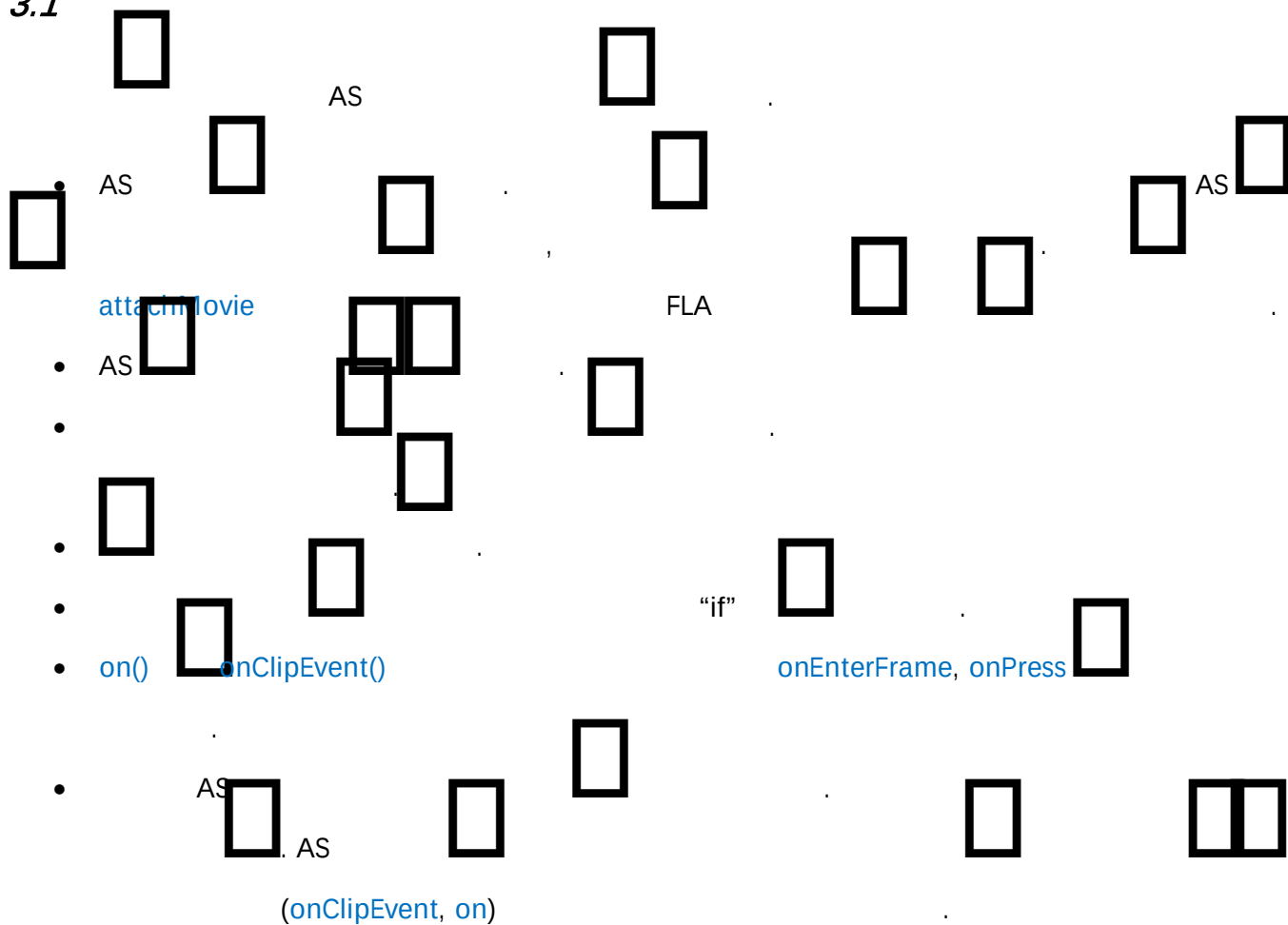




3



3.1



(gotoAndPlay, play, stop)

gotoAndPlay/gotoAndStop

o

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

o

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

gotoAndPlay/gotoAndStop

AS

[GFX::Movie::SetVariableArray](#)

scaleform

(Object)

- eval()

- Array.length

. myArr.length

```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
    trace(fontArr[i]);
}
```

```
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
    trace(fontArr[i]);
}
```

- (null

- removeListener()

- , AS VM

, AS if"

"com.xxx.yyy.aaa.bbb"

VM

"com", "xxx", "yyy", "aaa", "bbb"

/ yyy"

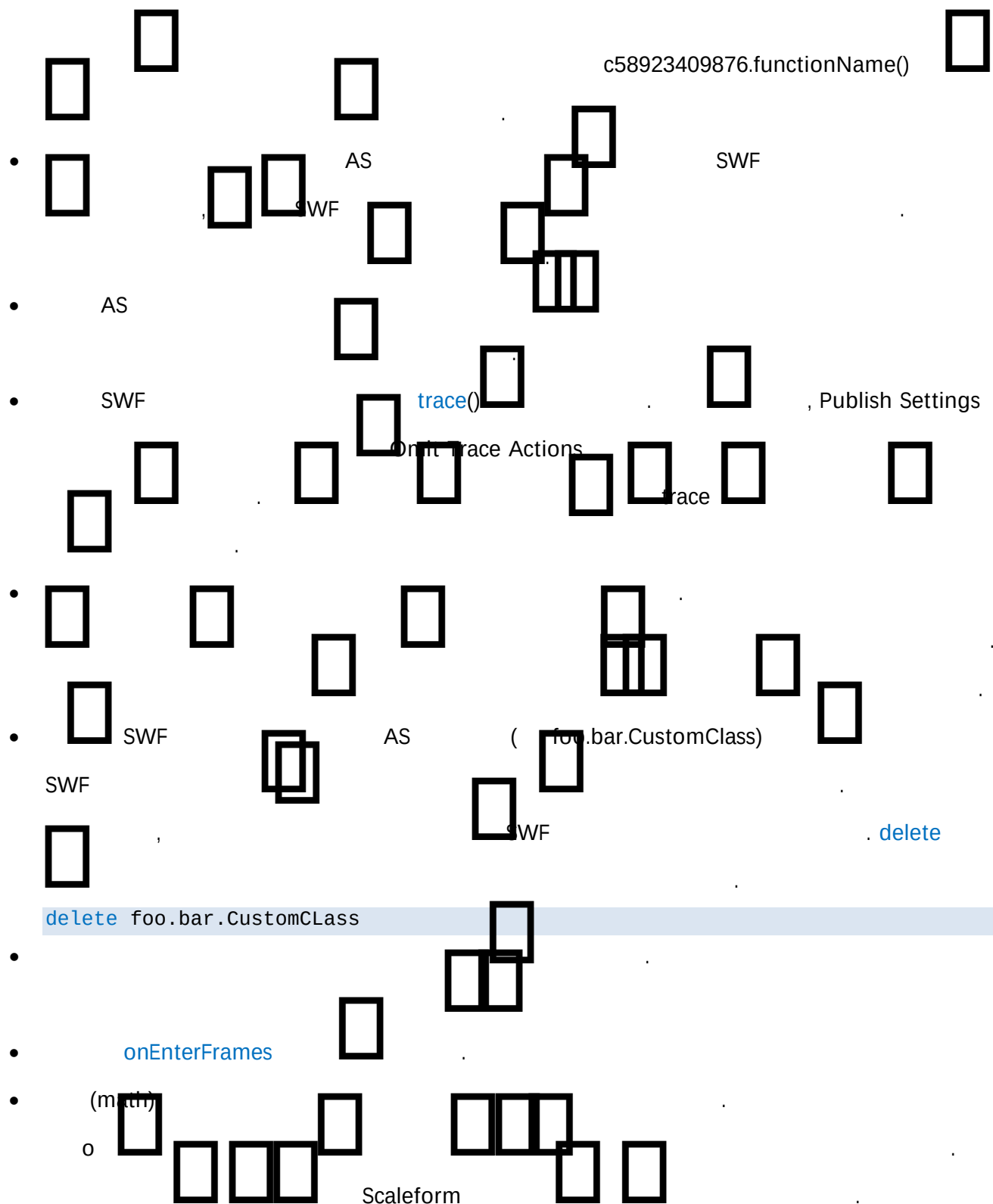
("com"

"xxx",

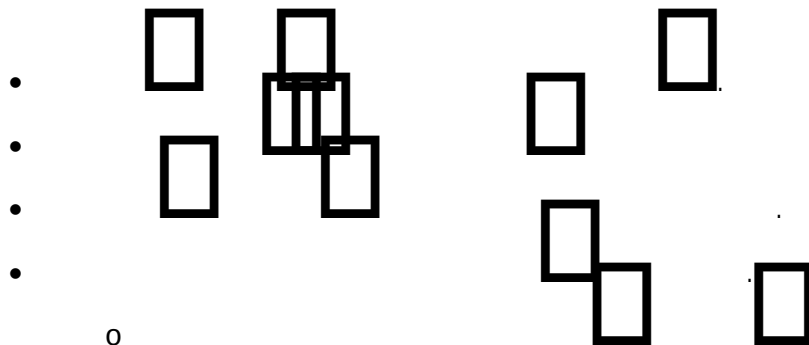
"xxx"

Flash

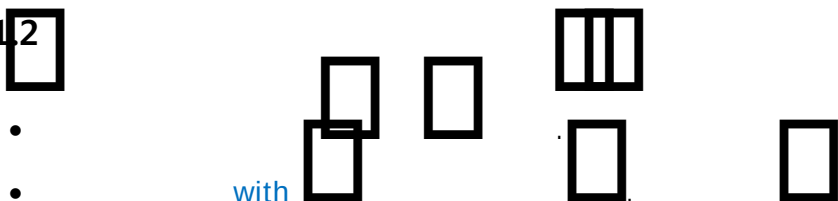
SWF



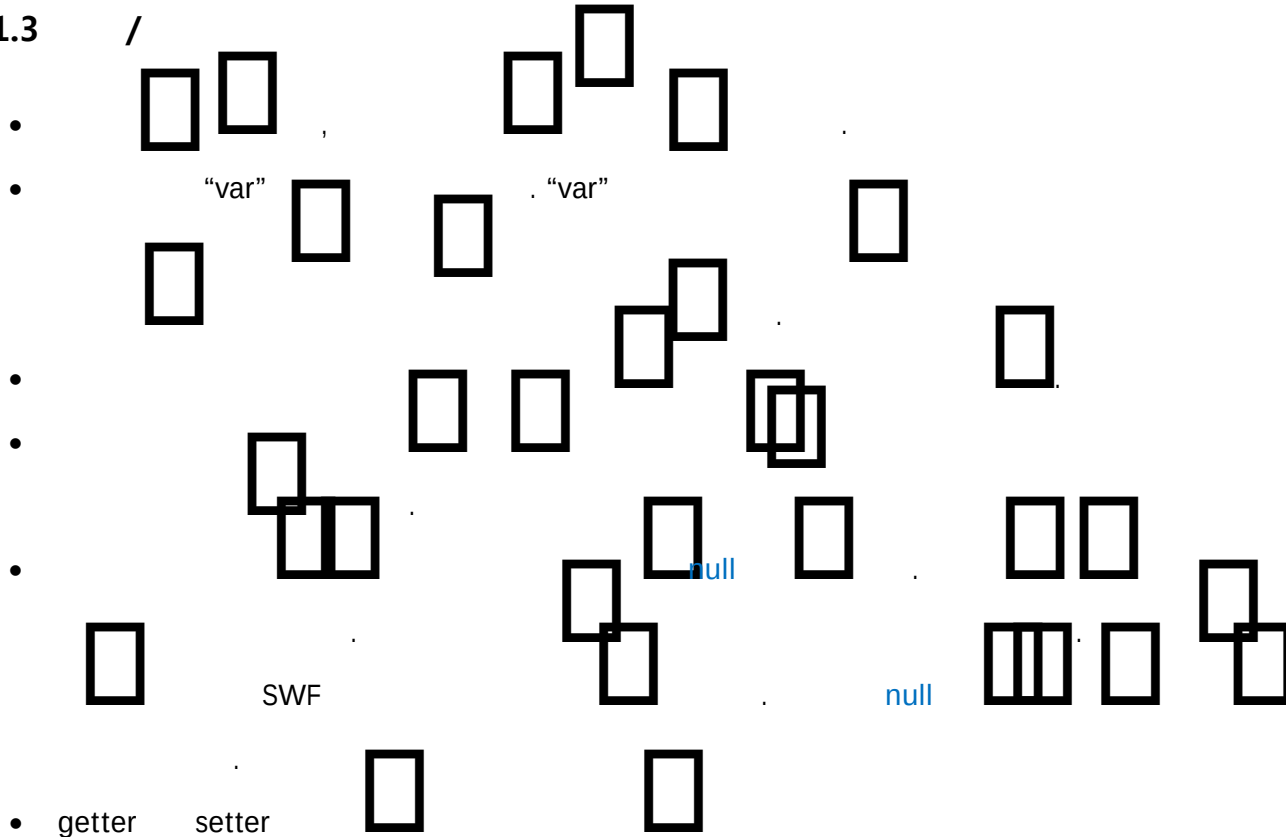
3.1.1



3.1.2



3.1.3





3.2 *ActionScript 3*

AS3



3.2.1



Object



. Object



. Object



AS3

- Array



4294967295(



Vector



)

Array

- Vector



. Vector<*>



. Vector<*>

Vector



Vector<int>, Vector<String>



Vector<YourFavoriteClass>



Vector<*>

- Object



flash.utils.Dictionary

-



GfX

Flash



-



(prototype)



AS2

. AS3



3.3 Advance

advance

1.

AS

onEnterFrame

/Event.ENTER_FRAME

2.

UI

3.

(_visible (AS2) or visible (AS3)

true

stop()

Advance

4.

_global.noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3)

Advance

Advance

"true"

_global.gfxExtensions (AS2) or scaleform.gfx.Extensions.enabled (AS3)

true

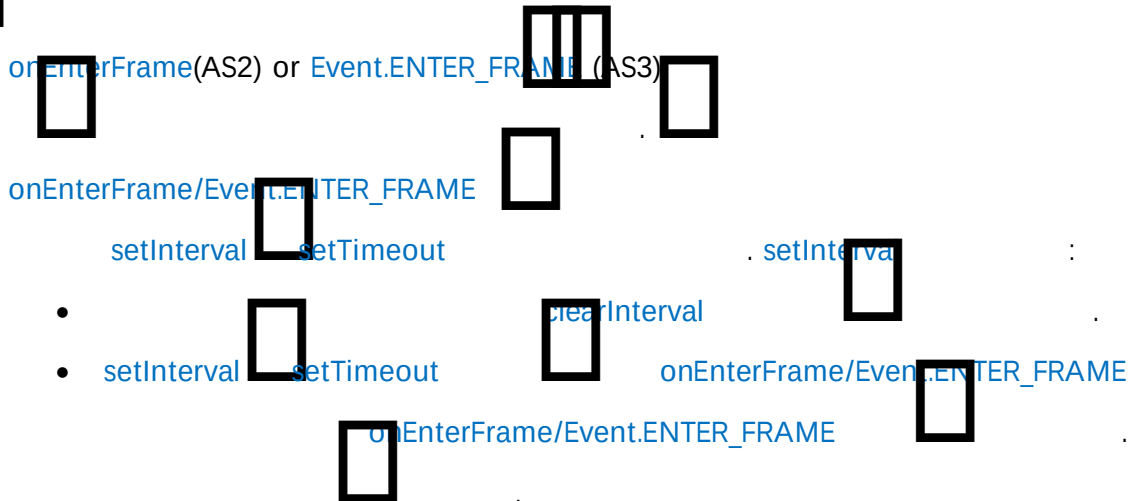
()

5.

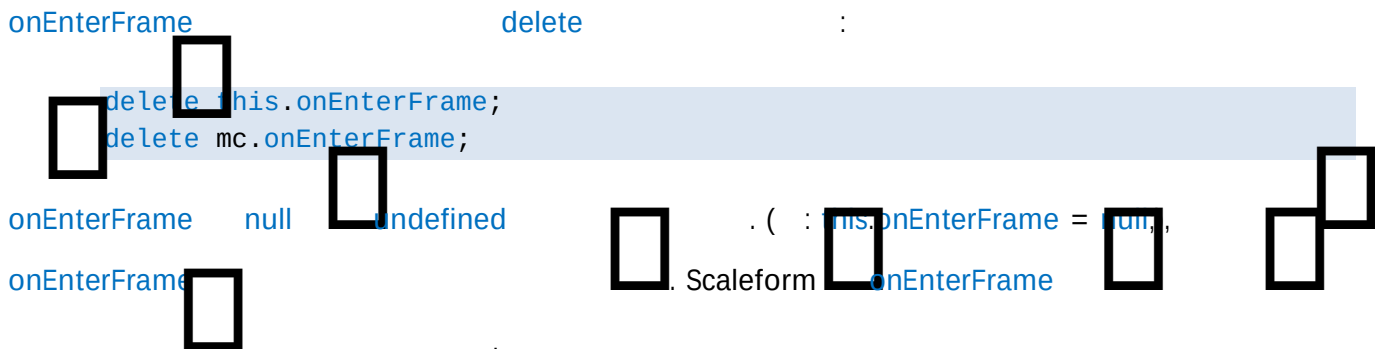
advance

6.

3.4 *onEnterFrame* and *Event.ENTER_FRAME*



3.4.1 *onEnterFrame*



3.5 *ActionScript 2*

3.5.1 *onClipEvent*



[illegible]

```
onClipEvent(construct)
{
    this.onLoad = function()
    {
        //function I
    }
}
```

onClipEvent(load)

3.5.2 Var



```
var i = 1000;
countIt = function()
{
    num = 0;
    for(j=0; j<i; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}
```

```

:
var i = 1000;
countIt = function()
{
    var num = 0;
    var ii = i;
    for(var j=0; j<ii; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}

```



3.5.3

(Precaching)



(var

),

```

function foo(var obj:Object)
{
    for (var i = 0; i < obj.size; i++)
    {
        obj.value[i] = obj.num1 * obj.num2;
    }
}

```

:

```

function foo(var obj:Object)
{
    var sz = obj.size;
    var n1 = obj.num1;
    var n2 = obj.num1;
    for (var i = 0; i < sz; i++)
    {
        obj.value[i] = n1*n2;
    }
}

```



```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
    // do something;
}
```

3.5.4



```
mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;
```



```
var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;
```

3.5.5



```
this[_global_.mynames[queue]][_global_.slots[i]].gosplash.text =
_global_.MyStrings[queue];
```

```
var _splqueue = this[_global_.mynames[queue]];
var _splstring = _global_.MyStrings[queue];
var slot_i = _global_.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;
```

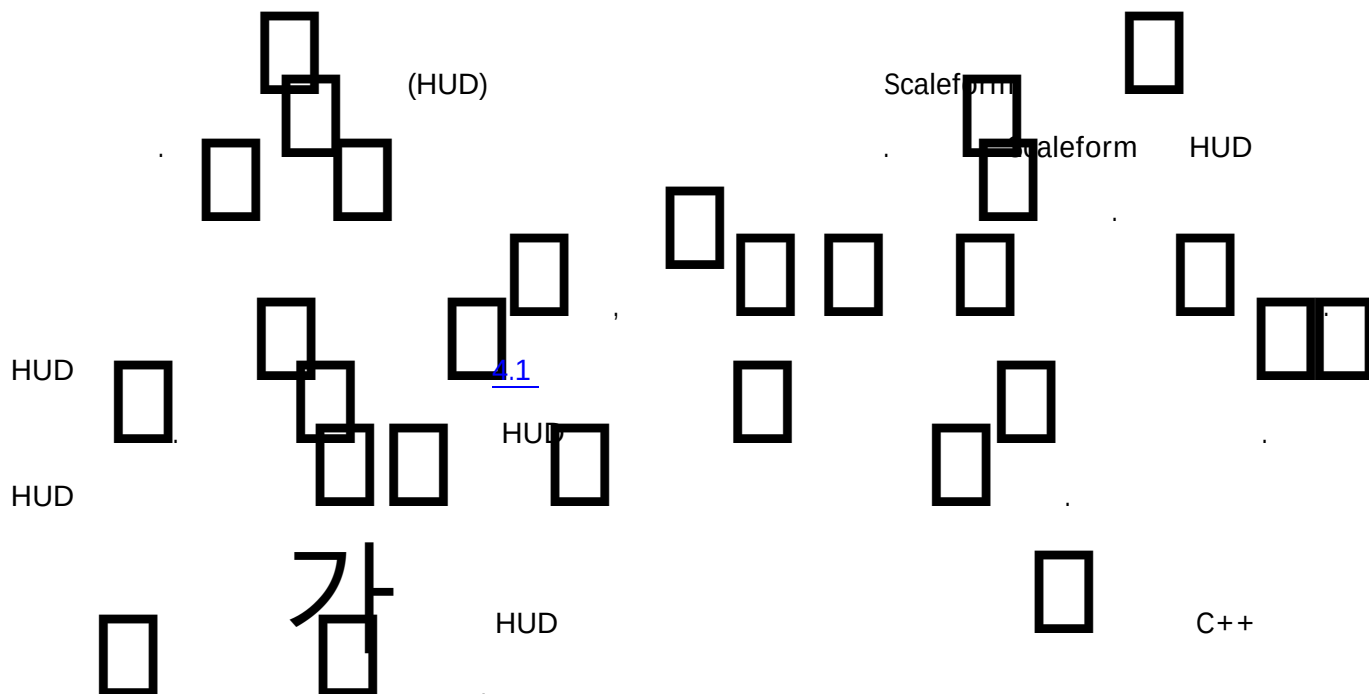


```
for(i=0; i<3; i++)
{
    this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
        _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
        _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
        0x000000;
}
```

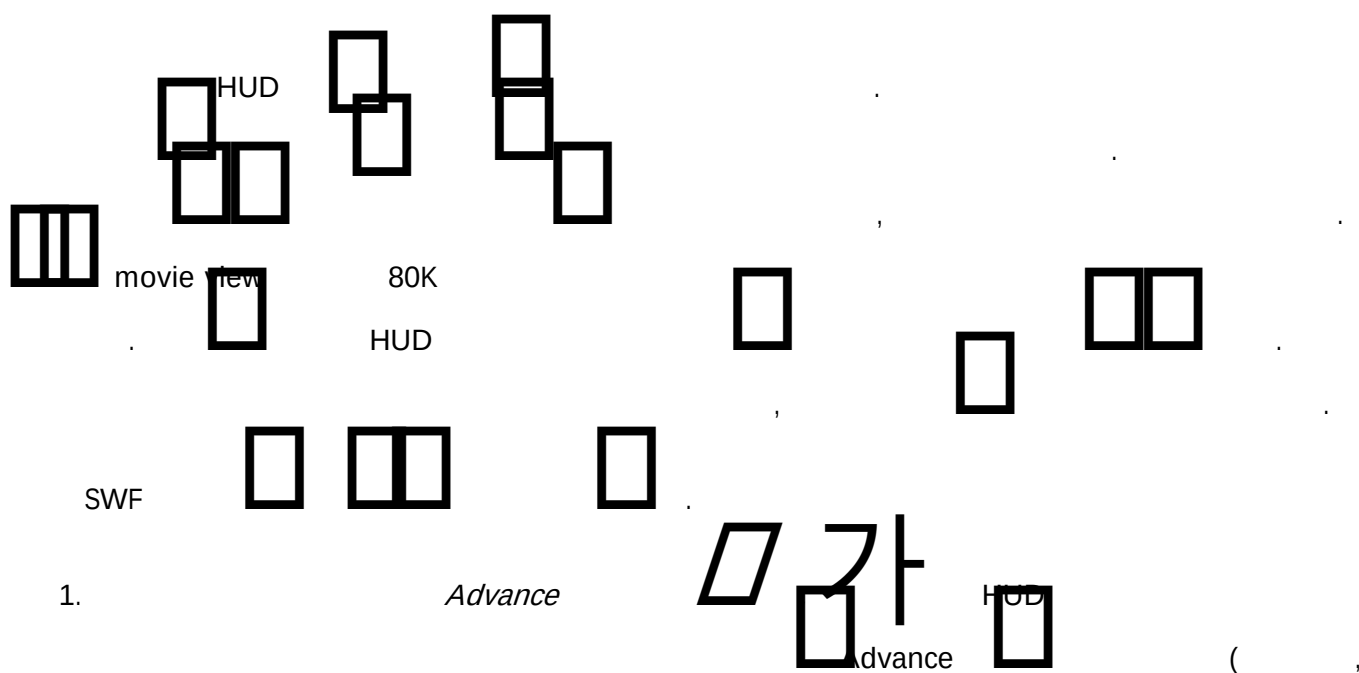
```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    _splqueue[slot_i].gosplash.text = _splstring;
    _splqueue[slot_i].gosplash2.text = splstring;
    _splqueue[slot_i].gosplash2.textColor = 0x000000;
}
```

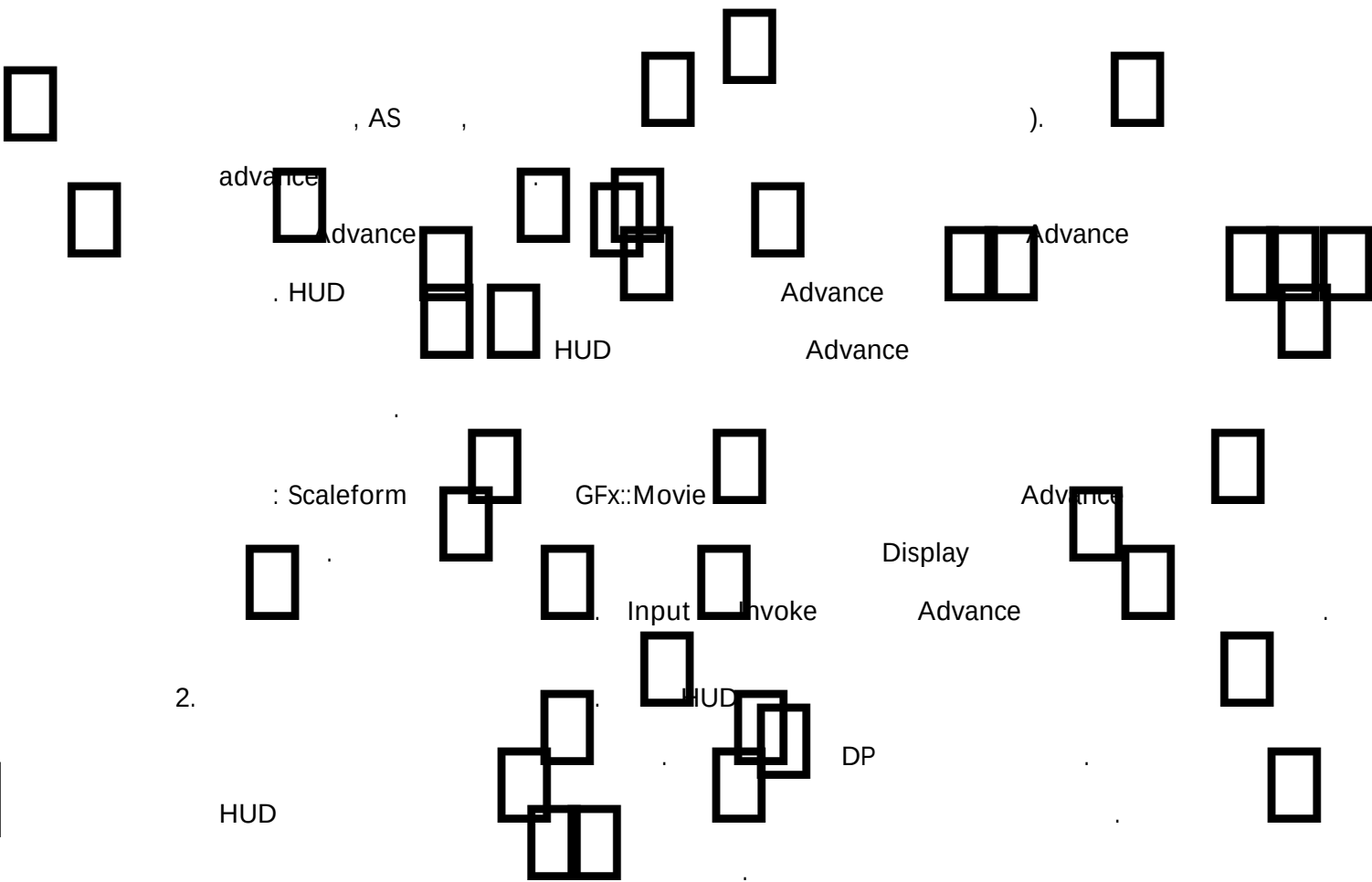


4 HUD



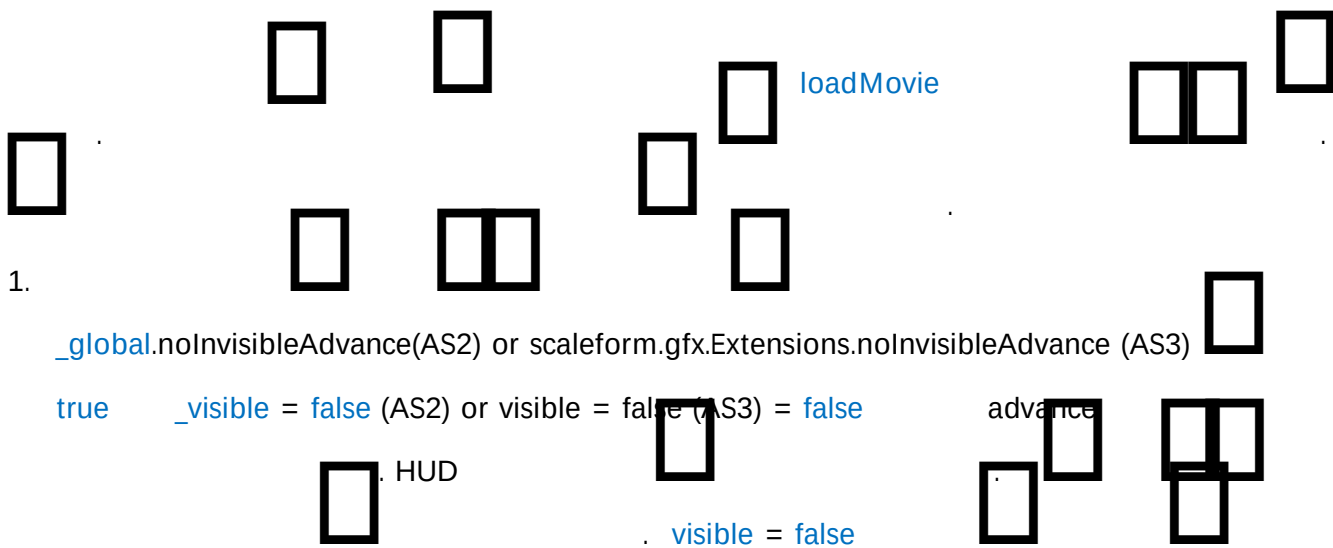
4.1 SWF





4.2

SWF



□



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HUD

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HUD



HUD 

C++

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Scaleform Player

advance HUD

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9



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DP

□

DrawText API

□

9



9

DI

7

, "Font

32

HUD

-
-

FAQ ["Graphics Rendering & Special Effects"](#)

- `Wii™` (
- `o GFX::Movie::SetMouseCursorCount(0,`
- `o`
- `HUD`

`noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3) \_visible`

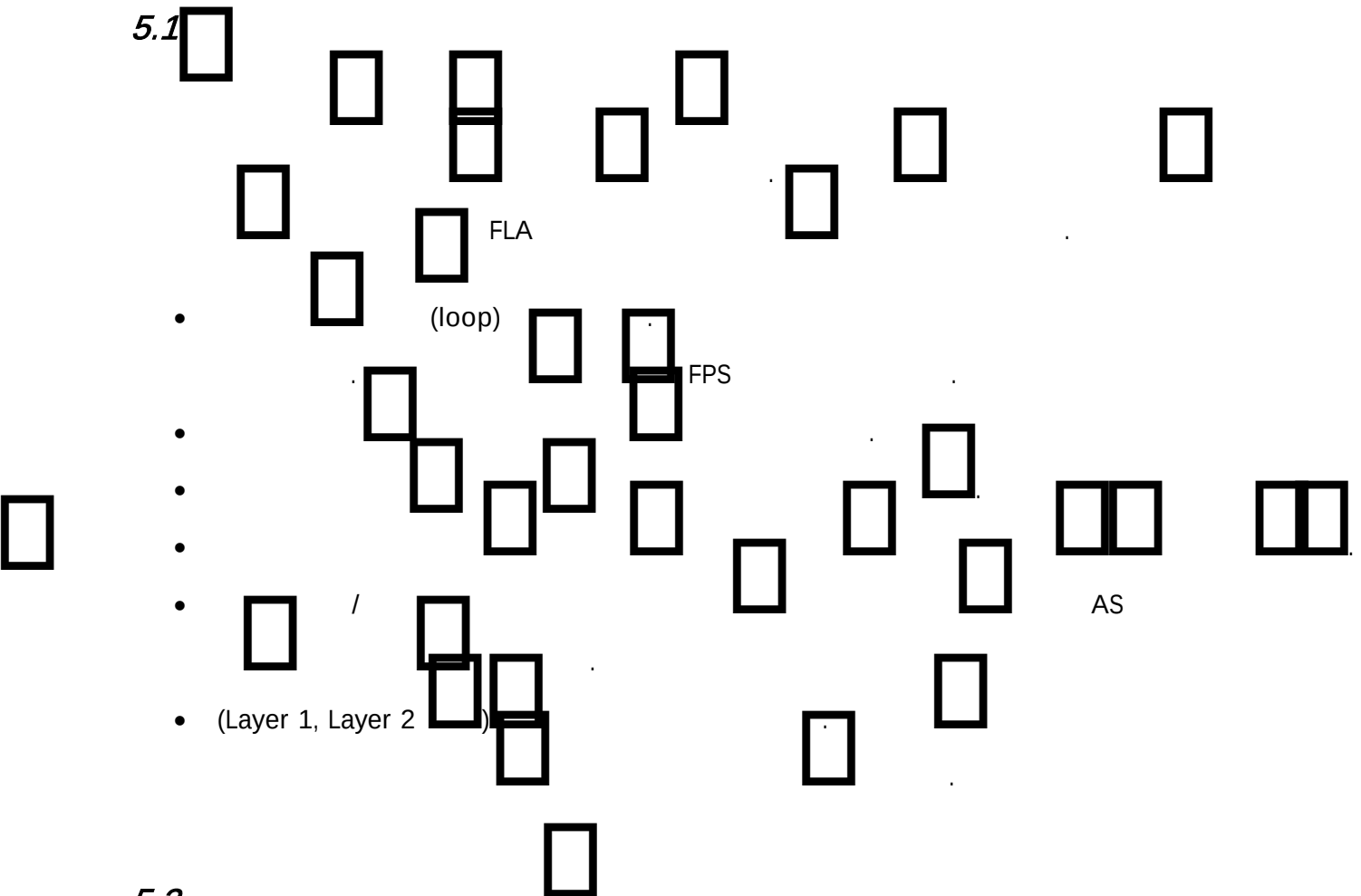
`(AS2) or visible (AS3)= false`

- `Invoke` . 2 (
- `Invoke` . ()
- `FAQ section`

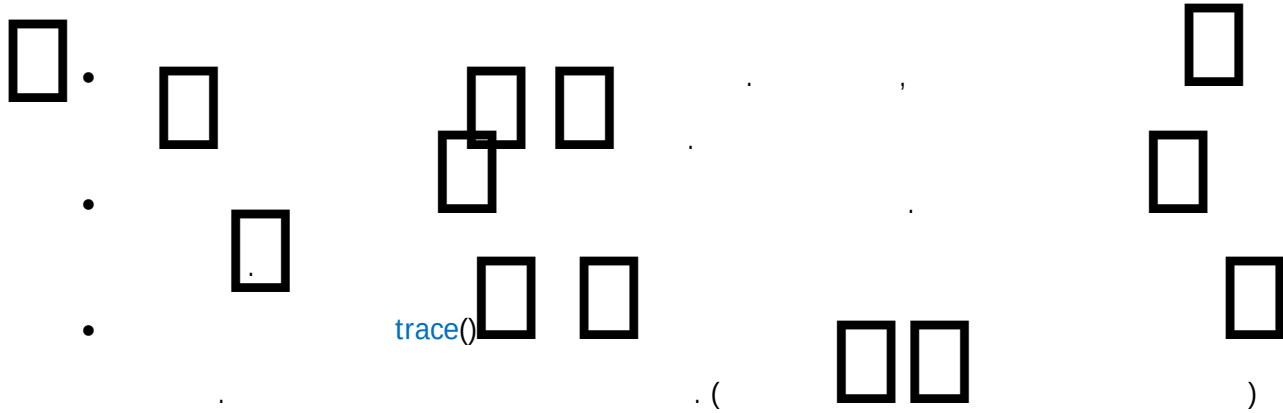
[2.1](#) ["Art & Assets"](#)

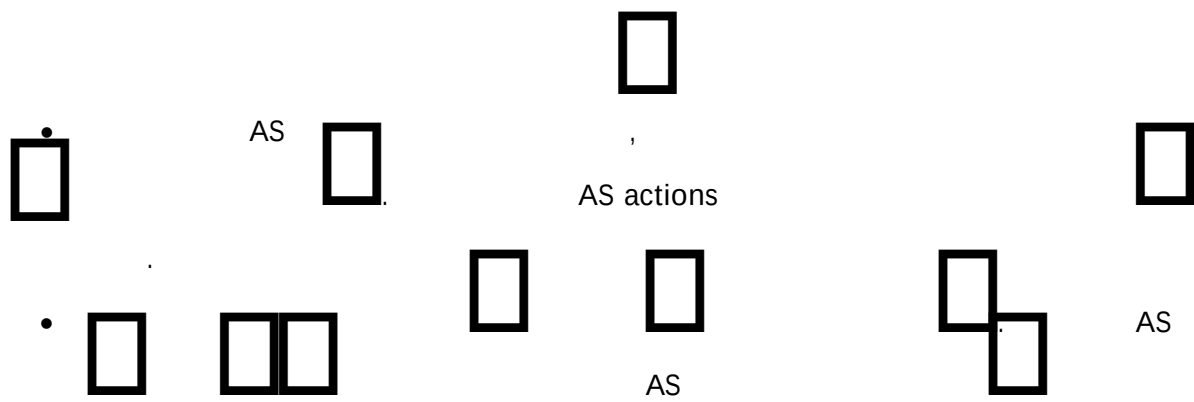
5

5.1



5.2





6

ActionScript 2.0 Best Practices

http://www.adobe.com/devnet/flash/articles/as_bestpractices.html

Flash 8 Best Practices

http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html

Flash ActionScript 2.0

http://www.adobe.com/devnet/flash/articles/actionscript_guide.html